



Decision Trees

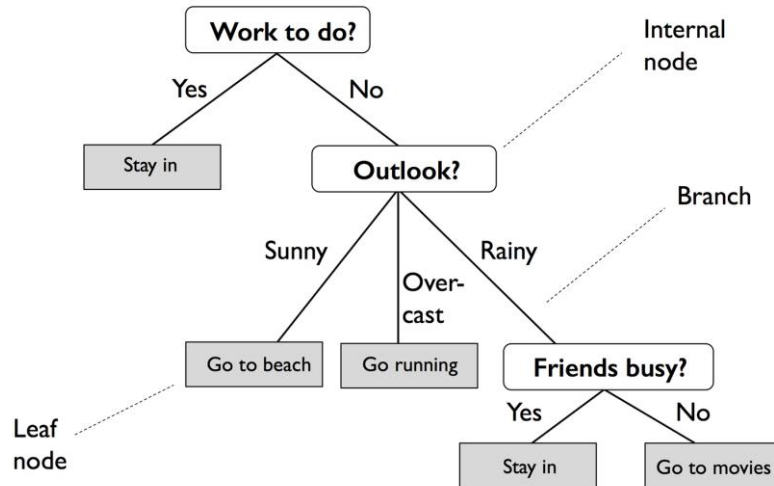
COMP7404

Computational Intelligence and Machine Learning

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Decision Tree Learning

- Decision tree classifiers are attractive models if we care about interpretability
- This model breaks down our data by making a decision based on asking a series of questions
- Example of decision tree



Decision Tree Learning

- Model learns to ask a series of questions
 - E.g.: Is sepal width ≥ 2.8 cm?
- Prediction is based on answers to questions
- What questions to ask?
 - Split data on feature that results in largest Information Gain (IG)
 - Iterative repeat splitting until leaves are pure
 - I.e., samples all belong to same class
 - This can result in a very deep tree with many nodes, which can easily lead to overfitting
 - Thus, we typically want to prune the tree by setting a limit for the maximal depth of the tree

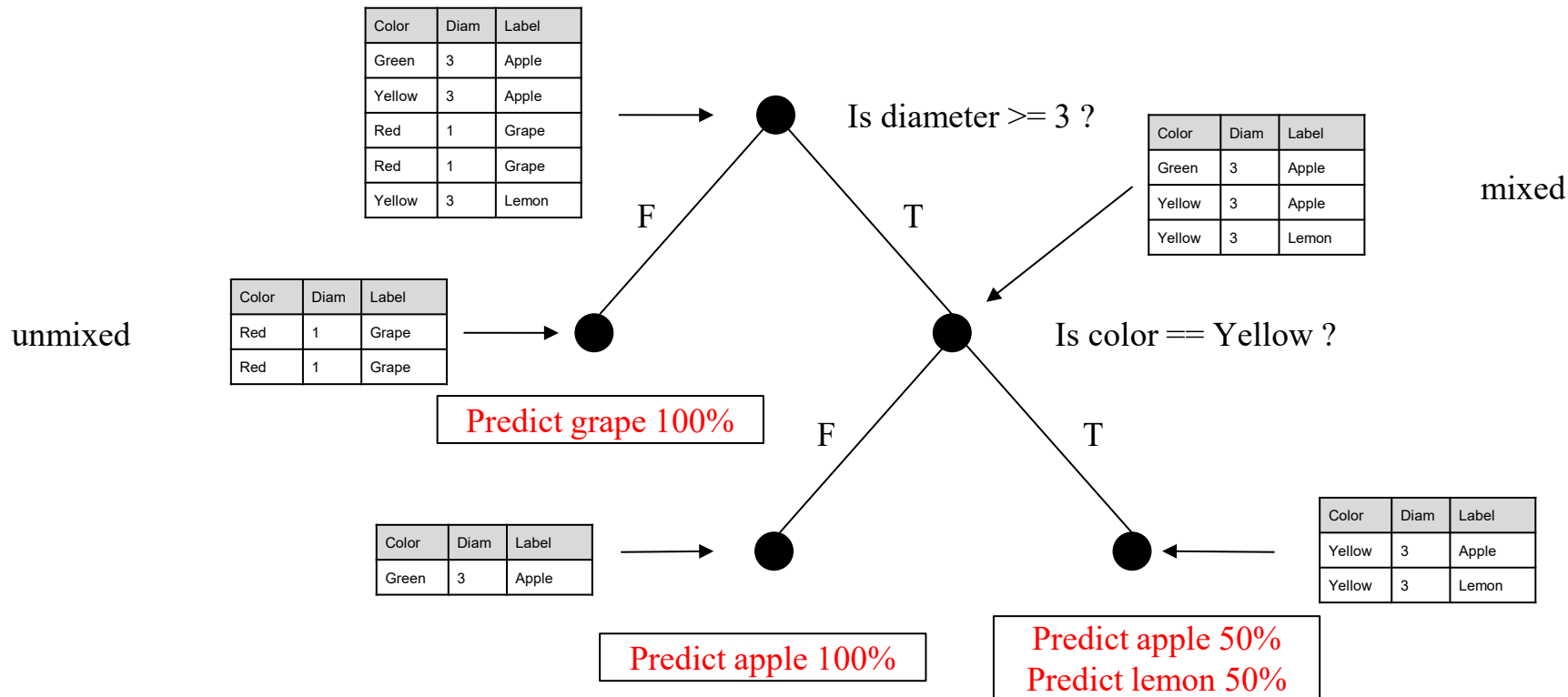
Decision Tree Learning - Intuition

- Training data:

Color	Diameter	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

features

Decision Tree Learning - Intuition



Which questions to ask and when?

- Quantify how much a question helps to unmix the labels
 - 1. Quantify the amount of uncertainty at a single node
 - E.g. Gini Impurity

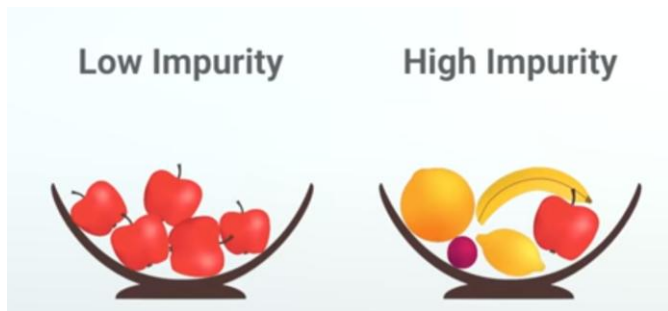
$$I_G(t)$$

- 2. How much does a question reduce the uncertainty
 - We aim to maximize Information Gain

$$IG(D_p, f)$$

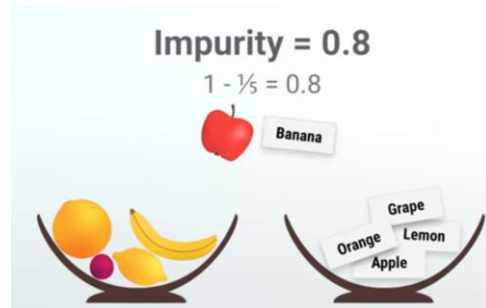
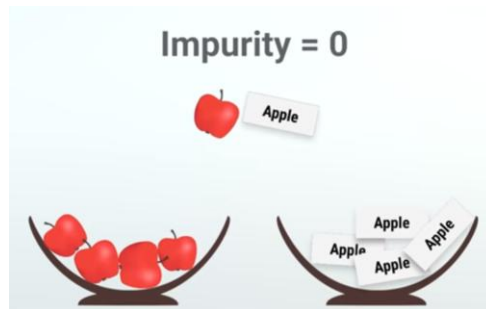
1. Gini Impurity

- Chance of being incorrect if you randomly assign a label to an example in the same set



$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$p(i|t)$ is the proportion of the samples that belong to class i for a particular node t



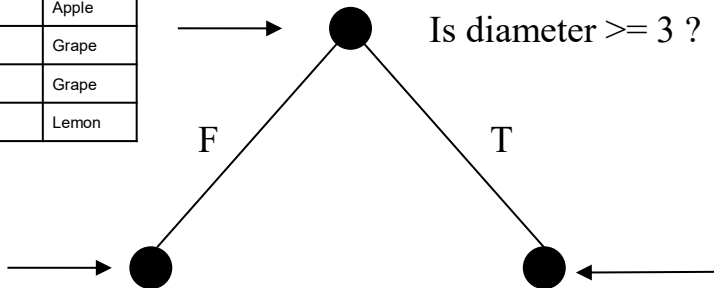
1. Gini Impurity

Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Gini Impurity:
 $I_G = 0$

Color	Diam	Label
Red	1	Grape
Red	1	Grape



Gini Impurity:
 $I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Yellow	3	Lemon

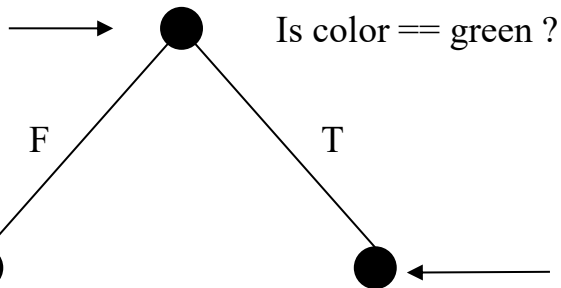
$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$p(i|t)$ is the proportion of the samples that belong to class i for a particular node t

1. Gini Impurity

Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon



Gini Impurity:
 $I_G = 0.625$

Color	Diam	Label
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Color	Diam	Label
Green	3	Apple

Gini Impurity:
 $I_G = 0$

$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$p(i|t)$ is the proportion of the samples that belong to class i for a particular node t

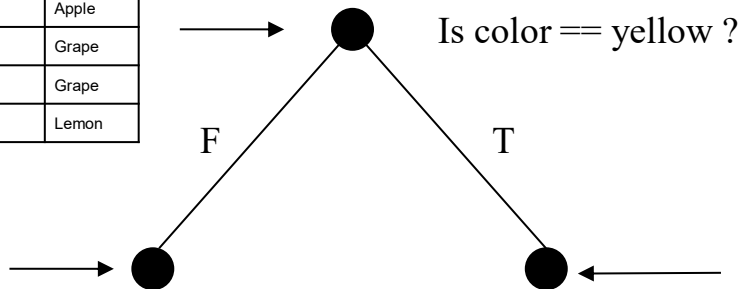
1. Gini Impurity

Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Gini Impurity:
 $I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Red	1	Grape
Red	1	Grape



Gini Impurity:
 $I_G = 0.5$

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon

$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$p(i|t)$ is the proportion of the samples that belong to class i for a particular node t

Which questions to ask and when?

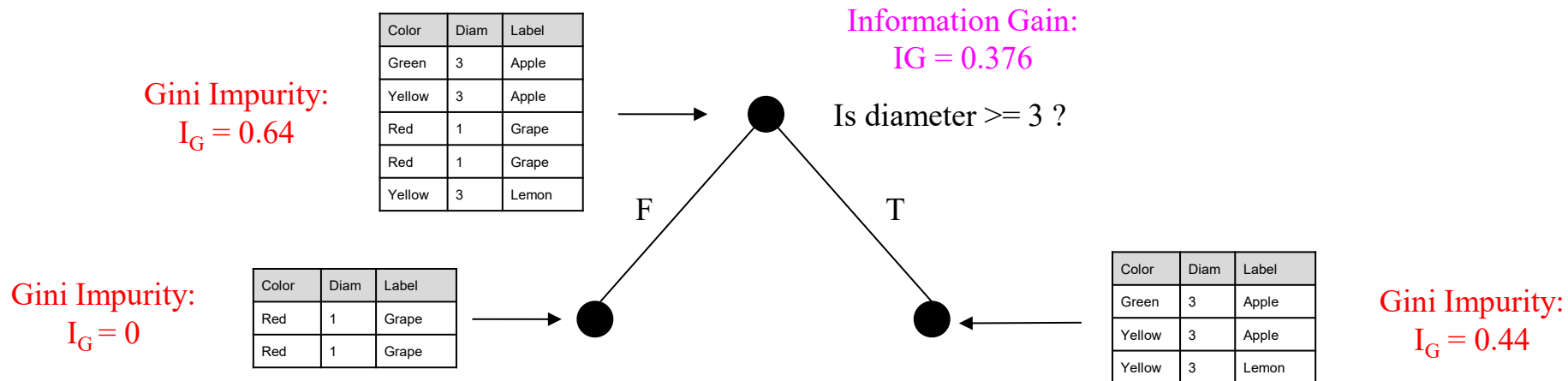
- Quantify how much a question helps to unmix the labels
 - 1. Quantify the amount of uncertainty at a single node
 - E.g. Gini Impurity

$$I_G(t)$$

- 2. How much does a question reduce the uncertainty
 - We aim to maximize Information Gain

$$IG(D_p, f)$$

2. Information Gain



$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

2. Information Gain - Task

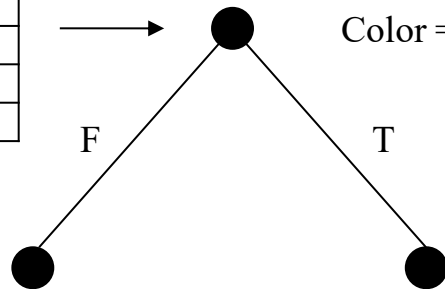
Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Information Gain:
 $IG = ?$

Color == green ?

Gini Impurity:
 $I_G = ?$



Gini Impurity:
 $I_G = ?$

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

2. Information Gain

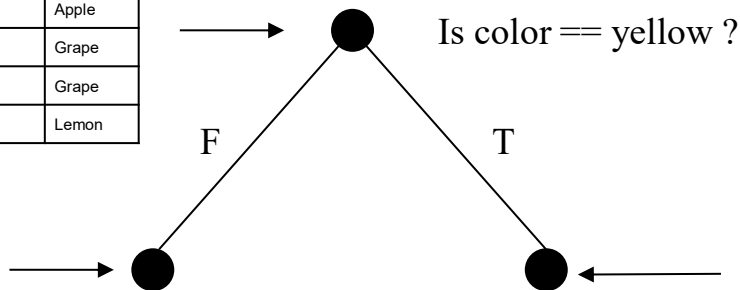
Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Gini Impurity:
 $I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Red	1	Grape
Red	1	Grape

Information Gain:
 $IG = 0.176$



Gini Impurity:
 $I_G = 0.5$

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

Example

$$I_G(t) = \sum_{i=1}^c p(i|t)(1-p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

$$IG(D_p, f) = I(D_p) - \frac{N_{left}}{N_p} I(D_{left}) - \frac{N_{right}}{N_p} I(D_{right})$$

Gini Impurity:
 $I_G = 0.64$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grape
Red	1	Grape
Yellow	3	Lemon

Gini Impurity:
 $I_G = 0$

Color	Diam	Label
Red	1	Grape
Red	1	Grape

Information Gain:
 $IG = 0.376$

Is diameter ≥ 3 ?

Gini Impurity:
 $I_G = 0.44$

Color	Diam	Label
Green	3	Apple
Yellow	3	Apple
Yellow	3	Lemon

Information Gain:
 $IG = 0.11$

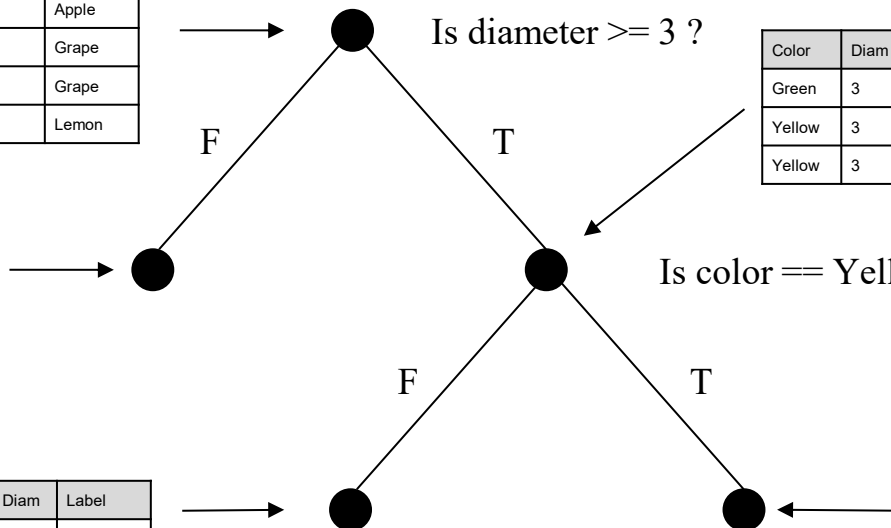
Is color == Yellow ?

Gini Impurity:
 $I_G = 0$

Color	Diam	Label
Green	3	Apple

Gini Impurity:
 $I_G = 0.5$

Color	Diam	Label
Yellow	3	Apple
Yellow	3	Lemon



Maximizing Information Gain

- Split nodes at most informative features
 - In our tree learning algorithm we optimize an objective function
 - E.g., maximize information gain at each split

$$IG(D_p, f) = I(D_p) - \sum_{j=1}^m \frac{N_j}{N_p} I(D_j)$$

- f is the feature to perform the split
- D_p and D_j are the dataset of the parent and j th child node
- I is our impurity measure
- N_p is the total number of samples at the parent node, and
- N_j is the number of samples in the j th child node.

Binary Decision Tree

- For binary decision trees each parent node is split into two child nodes
 - D_{left} and D_{right}

$$IG(D_p, f) = I(D_p) - \frac{N_{\text{left}}}{N_p} I(D_{\text{left}}) - \frac{N_{\text{right}}}{N_p} I(D_{\text{right}})$$

- The following impurity measures or splitting criteria are commonly used in binary decision trees
 - Gini impurity (I_G),
 - Entropy (I_H), and
 - Classification error (I_E)

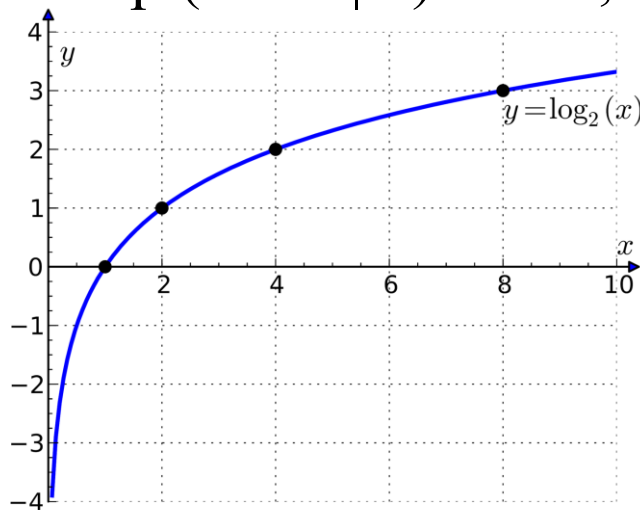
Entropy

$$I_H(t) = - \sum_{i=1}^c p(i|t) \log_2 p(i|t)$$

- $p(i|t)$ is the proportion of the samples that belong to class i for a particular node t
- The entropy is therefore 0 if all samples at a node belong to the same class, and the entropy is maximal if we have a uniform class distribution
- The entropy criterion attempts to maximize the mutual information in the tree

Entropy

- For example, in a binary class setting ($c = 2$)
 - The entropy is 0 if $p(i = 1 | t) = 1$ or $p(i = 0 | t) = 0$
 - If the classes are distributed uniformly with $p(i = 1 | t) = 0.5$ and $p(i = 0 | t) = 0.5$, the entropy is 1



$$I_H(t) = -\sum_{i=1}^c p(i|t) \log_2 p(i|t)$$

$p(i | t)$ is the proportion of the samples that belong to class i for a particular node t

Gini

- The Gini impurity can be understood as a criterion to minimize the probability of misclassification

$$I_G(t) = \sum_{i=1}^c p(i|t)(1 - p(i|t)) = 1 - \sum_{i=1}^c p(i|t)^2$$

- Similar to entropy, the Gini impurity is maximal if the classes are perfectly mixed, for example, in a binary class setting ($c = 2$)

$$I_G(t) = 1 - \sum_{i=1}^c 0.5^2 = 0.5$$

Classification Error

- Another impurity measure is the classification error

$$I_E = 1 - \max \{ p(i | t) \}$$

- This is a useful criterion for pruning but not recommended for growing a decision tree, since it is less sensitive to changes in the class probabilities of the nodes

Code - DecisionTrees.ipynb

- Available [here](#) on CoLab

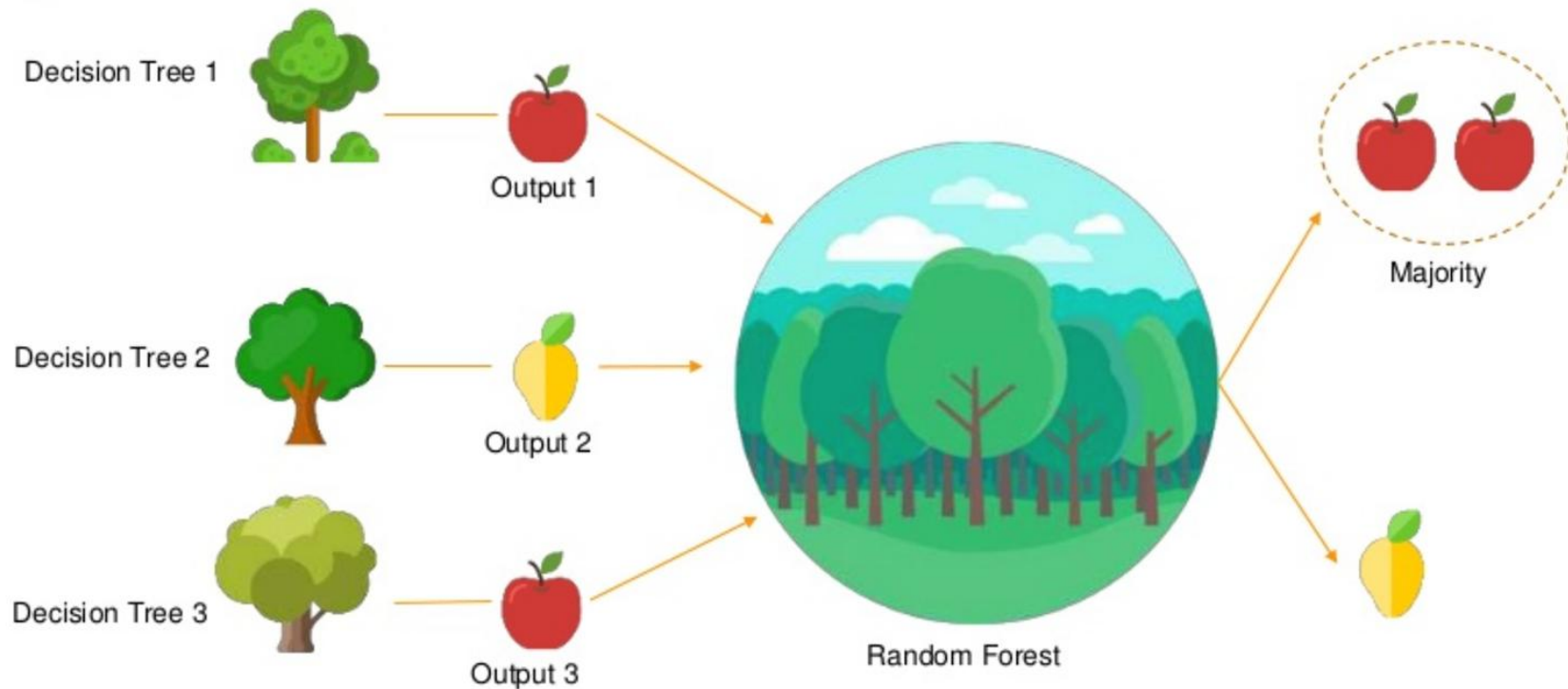
Random Forests

- A random forest can be considered as an ensemble of decision trees
- The idea behind a random forest is to average multiple (deep) decision trees that individually suffer from high variance, to build a more robust model that has a better generalization performance and is less susceptible to overfitting

Random Forests Creation

1. Draw a random bootstrap sample of size n (i.e., randomly choose n samples from the training set with replacement)
2. Grow a decision tree from the bootstrap sample. At each node:
 - a. Randomly select d features without replacement
 - b. Split the node using the feature that provides the best split according to the objective function, for instance, maximizing the information gain
3. Repeat steps 1. and 2. k times
4. Aggregate the prediction by each tree to assign the class label by majority vote

Random Forests Intuition



Bootstrap Sample Size

- Decreasing the size of the bootstrap samples (i.e., decreasing n) increases the diversity among the individual trees
 - The probability that a particular training sample is included in the bootstrap sample is lower
 - This increases the randomness of the random forest
 - Helps to reduce the effect of overfitting
- However, smaller bootstrap samples typically result in a lower overall performance of the random forest, a small gap between training and test performance, but a low test performance overall
- Conversely, increasing the size of the bootstrap sample may increase the degree of overfitting
 - The bootstrap samples, and consequently the individual decision trees, become more similar to each other, they learn to fit the original training dataset more closely